

Sentiment Analysis on Product Reviews

Objective:

In this project, I aimed to analyse customer reviews and understand what customers actually feel about the products. Instead of relying only on ratings, I used sentiment analysis to classify reviews as Positive, Negative, or Neutral and extract meaningful insights.

Dataset & Tools Used:

For this analysis, I used the Amazon Fine Food Reviews dataset and worked with the first 5000 rows to keep the project manageable. The tools I used include Python, Pandas for data handling, Text Blob for sentiment analysis, and Matplotlib/Seaborn for creating visualizations.

Approach:

I started by loading and exploring the dataset to understand its structure. After that, I cleaned the data by removing null values, empty reviews, and duplicate entries, and kept only the required columns (Text and Score). Then, I used Text Blob to calculate the sentiment polarity of each review and classified them into Positive, Negative, or Neutral categories. Finally, I created different charts to better visualize and understand the results.

Key Findings:

After analyzing the data, I can say that around **88.34% of the reviews are positive**, which clearly shows that most customers are satisfied with the products. From my observation, negative reviews are much fewer and are mostly related to issues like product quality, taste, and sometimes packaging.

Chart 1

This bar chart shows the number of reviews in each sentiment category. We can clearly see that positive reviews are significantly higher compared to negative and neutral ones, which indicates overall customer satisfaction.



Chart 2

This pie chart represents the percentage distribution of sentiments. A large portion of the chart is occupied by positive reviews, confirming that most customers have a favorable opinion about the products.

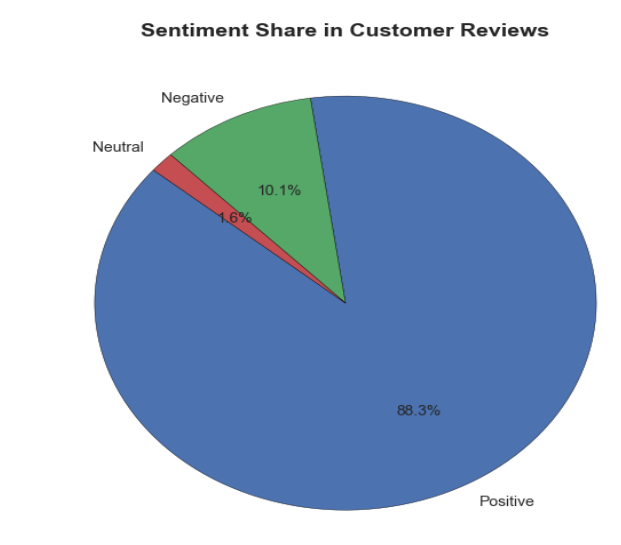


Chart 3

This chart compares product ratings with the detected sentiment. In most cases, higher ratings align with positive sentiment, but there are a few mismatches, showing that customer ratings and written reviews do not always perfectly match.



Conclusion:

From my analysis, I can say that sentiment analysis provides a deeper understanding of customer opinions compared to just ratings. It helps identify what customers actually feel and where improvements are needed. Based on the findings, focusing on improving product quality and packaging can help reduce negative feedback and further improve customer satisfaction.